How Alcohol and Other Substances Affect the Brain and Body

Category 01: Understanding Alcohol & Drugs · Topic 02

A complete educational learning module for young people, adults, and families. Includes accessible science, practical examples, fictional interactive activities, health-safety guidance, and traceable references.

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AUDIENCE · YOUTH / ADULTS / CAREGIVERS · NO AGE GATE

PUBLICATION STATUS · REQUIRES CLINICAL, YOUTH-SAFEGUARDING, AND LOCALIZATION REVIEW

A PDF can describe interaction scripts and feedback, but the selectable options will only become functional when the website is implemented. No personal screening or disclosure is required.

How to use this module

- Read the main explanation in short sections; skip to the safety section whenever needed.
- Choose optional fictional activities to practice understanding. There are no diagnostic scores.
- Access the essential information in text even without video, audio, or an internet connection.

01 | Editorial review of Topic 01

Review scope: Clarity, educational structure and learning-activity design, using the full text retrieved from the user’s Google Drive file on 11 November 2026. This is an editorial review, not a formal clinical audit.

Strengths to retain

- Definitions of alcohol, medicine, drug and psychoactive substance are distinct and understandable.
- Fictional examples keep learning available to adolescents without demanding personal disclosure.
- Activities explain the reasoning behind safer answers instead of just declaring right and wrong.
- The safety section appropriately flags intoxication, emergencies and dangerous alcohol withdrawal.

Improvements applied in Topic 02

- Separate immediate effects, repeated-use consequences and substance use disorders rather than implying they are the same.
- Explain brain functions in plain terms without oversimplifying addiction into a single chemical.
- Make each activity accessible through an identical read-only answer-and-feedback pathway in the PDF.
- Make sure emergency advice is usable without presenting US emergency numbers as Rwanda contacts.

Review still outstanding

A qualified clinician should validate medical statements. A youth safeguarding reviewer, local policy reviewer, and Kinyarwanda/French language reviewers should approve the final content before release.

02 | Start with the big idea

Learning objectives: After this topic, a learner should be able to explain how substances reach and affect the brain; identify key body systems that can be affected; distinguish short-term effects from long-term harm; recognize risk differences; and identify symptoms needing urgent assistance.

Alcohol and other psychoactive substances affect the body because their molecules interact with cells and biological systems. Many substances influence the brain, changing alertness, judgment, memory, movement, mood or pain perception. They may also affect breathing, the heart, liver, digestive system and other organs. Effects depend on the particular substance and the person’s circumstances. [1-5]

Three ideas to remember

- Immediate effect: what changes while a substance is acting in the body (for example, slower reactions).
- Longer-term risk: what may happen through repeated exposure or certain patterns of use (for example, liver disease).
- Substance use disorder: a diagnosed health condition involving a harmful pattern of use; it does not occur in everyone who uses a substance.

The risk is not the same for every substance, dose, person, or setting. This lesson describes patterns rather than predicting what will happen to an individual.

Everyday example

A fictional adult drinks alcohol at a celebration and feels confident and talkative. Their sense of confidence does not prove that their balance, reaction time, or driving ability is intact. This illustrates how subjective feelings and actual performance can differ.

03 | What happens in the brain?

A short explanation of nerve-cell communication

The brain consists of billions of nerve cells, called neurons. They communicate using electrical activity and chemical messengers called neurotransmitters. Substances can change how chemical signals are released, received, or cleared. These changes can influence attention, emotions, pain, movement, sleep, memory and behavior. [2,3]

Five brain functions worth understanding

Brain function	Plain-language explanation
Judgment and self-control	Helps a person consider consequences, resist impulses and plan actions.
Memory and learning	Helps encode experiences and remember new information.
Coordination	Helps regulate movement and balance.
Reward and motivation	Helps us learn which actions are important and worth repeating.
Alertness and breathing control	Helps regulate wakefulness and essential automatic functions.

What about dopamine?

Dopamine helps the brain learn from rewards and cues. Some substances strongly reinforce repeated use through reward-related circuits. However, addiction is not simply “too much dopamine.” It involves multiple brain systems, learning processes, environment, biology and personal circumstances. [3,4]

A helpful analogy: the brain is a network of cooperating systems, not a single on/off switch. That is why two substances, or two people using the same substance, may have different effects.

04 | Alcohol: immediate effects and longer-term risks

What happens after drinking?

Alcohol contains ethanol, a psychoactive and toxic substance. It can interfere with neural communication and affect balance, memory, speech, judgment and reaction time. At high exposure, alcohol poisoning can impair breathing and consciousness. Its effects are not reliably judged by how confident or “normal” someone feels. [1,2]

Different body systems

System	Potential effects or health concerns
Brain and nerves	Impaired attention, memory, motor coordination; long-term heavy use can damage nervous tissue.
Liver and digestion	Risk of liver disease and digestive or pancreatic conditions.
Heart and circulation	High blood pressure, irregular heart rhythm and other cardiovascular conditions.
Cancer risk	Alcohol increases risk for several cancers, including breast, liver and colorectal cancers.
Relationships and safety	Injury, interpersonal harm, road crashes, school or work difficulties.

The WHO states that no pattern of alcohol consumption is completely risk-free. Risk varies with exposure and personal circumstances; many harms are concentrated in heavy episodic or sustained heavy drinking. [1]

Misconception check

“I am not dependent, so alcohol cannot harm me.” Incorrect. Injuries, impaired judgment and some health risks can occur without dependence. [1]

05 | How different drugs act differently

The term “drugs” covers many different kinds of substances. Medicines can be useful when prescribed for an appropriate condition, but using another person’s medicine or combining substances can be dangerous. The categories below are broad and not a guide to consumption. [3-5]

Group	Possible effects	Important concerns
Opioids	Pain relief and sedation	Breathing may slow; overdose can be fatal.
Stimulants	Increased arousal and alertness	Possible agitation, temperature and cardiovascular problems.
Sedatives	Drowsiness and slowed responses	Impaired coordination; dangerous with alcohol or opioids.
Cannabis	Changes in attention and perception	Memory, coordination and mental health concerns vary by age and context.
Nicotine	Stimulation and reinforcement	Dependence and significant health risks, especially with tobacco.
Inhalants	Dizziness, altered awareness	Can cause sudden and severe toxic emergencies.

Why mixing substances matters

Some effects interact. For example, alcohol used together with certain sedatives or opioids may increase the risk of dangerous breathing suppression. Someone should ask a qualified clinician or pharmacist about prescribed medicines rather than relying on social advice or a general learning page. [5]

06 | Why age and life circumstances matter

Young people and the developing brain

Brain maturation continues through adolescence and into young adulthood. Alcohol exposure during this period may affect developing systems involved in decision-making, learning and memory. Earlier use and heavy patterns are associated with additional harms, but an individual outcome cannot be predicted from age alone. [2,6]

Adults and older people

Risk also matters later in life. Medical conditions, medicines, work demands, pregnancy and other circumstances may change how a substance affects someone. In older adults, medication interactions and fall risk can be particularly relevant. A person’s vulnerability should not be judged solely by appearance. [1,5]

What influences risk?

- The substance, its potency, exposure and combinations with other substances.
- Age, pregnancy, underlying illness, prescribed medicines and sleep.
- Stress, trauma, social support, access to care and environment.
- Frequency and pattern of use, as well as whether impairment occurs in dangerous situations.

Fictional Rwanda-relevant situations

Secondary school: A learner is told alcohol helps people become confident in public. A safe learning response is to explain that temporary feelings can coexist with impaired judgment.

Musanze transport: A motorcyclist has been drinking and feels alert. The immediate safety lesson is not to drive or ride while impaired.

Work and medicines: An adult with a prescribed pain medicine plans to drink. The person should check the interaction with their clinician or pharmacist.

07 | Four real-life learning cases

Case A — Memory and examinations

A fictional 19-year-old student reports poor recall after drinking heavily at a gathering. Alcohol can disrupt the formation of memories. The lesson is that missing memories are not harmless signs of a “good time” and may indicate significant intoxication. [6]

Case B — A family member who seems overly sleepy

An adult becomes difficult to awaken after taking an unknown substance. Friends assume the person should sleep it off. Unresponsiveness or unusual breathing may signal a medical emergency; urgent care is more appropriate than waiting. [7]

Case C — A prescribed medicine shared with someone

A friend offers another person a sedative prescribed to them, saying it is safe because a doctor issued it. Prescriptions are specific to the individual and their condition. Sharing medicines can introduce dangerous interactions and should be avoided.

Case D — Long-term heavy drinking

A worker describes worsening sleep, health and family functioning after a prolonged pattern of heavy drinking. A health professional can assess physical and psychological needs; a short online lesson cannot diagnose alcohol use disorder or decide whether stopping abruptly is medically safe.

Group discussion: For each case, ask what is known, what cannot be assumed, and what safer next action is possible. Never require participants to share their own substance-use history.

08 | Interactive learning activities

These choices and feedback are scripts for future website interactions. In this PDF, the answer and explanation are printed beneath each scenario. These activities are not screening or diagnosis.

Activity 1 — Brain and behavior

A person feels confident after alcohol but finds concentration difficult. Which statement is best?

- A. Confidence proves the person is safe to drive.
- B. Alcohol may impair judgment even if the person feels confident.
- C. Alcohol only affects the liver.

Feedback: B. Alcohol can affect brain processes underlying attention, reaction time and judgment. [2]

Activity 2 — Substances and risk

Which statement is accurate? A. Every psychoactive medicine is harmful when appropriately used. B. All substances have identical effects. C. Effects and risks depend on the substance and person.

Feedback: C. Medicines may be beneficial in appropriate care, and different substances act through different pathways.

Activity 3 — Emergency recognition

A fictional person cannot be awakened and has slow, abnormal breathing. Which response is safest? A. Let them sleep. B. Seek emergency medical help immediately. C. Give them coffee.

Feedback: B. These can be signs of life-threatening poisoning or overdose. Do not delay urgent care. [7]

Activity 4 — Adolescents

Why do adolescents need particular protection? A. Brain development is ongoing. B. They have no brain development left. C. Only adults can experience harm.

Feedback: A. Brain development continues into young adulthood, and substance-related harms can affect learning and safety. [6]

09 | Safe actions, help-seeking and takeaways

When urgent assistance matters

Seek immediate medical help for unresponsiveness, slow or absent breathing, seizures, collapse, or other severe signs of poisoning. Contact the relevant local emergency service or go to the nearest emergency department. Follow professional dispatcher instructions. Do not leave an unresponsive person alone while help is being arranged. [7]

For a suspected opioid overdose, naloxone can reverse opioid effects if available and given by someone trained or following authorized instructions. Emergency medical care is still necessary. Local availability should be verified before publication. [7]

Important warning about withdrawal

Someone with possible alcohol dependence should obtain clinical advice before suddenly stopping heavy prolonged use: withdrawal can cause serious complications and may require supervised care. This topic does not teach detoxification or prescribe medication.

Key takeaways

- Alcohol and other psychoactive substances can affect several brain and body systems.
- Short-term impairment, long-term health risks and substance use disorders are different concepts.
- Alcohol can impair judgment and coordination even without obvious outward signs.
- Substances vary in mechanism and risk; mixing them may increase danger.
- Growing brains and particular health circumstances can require additional protection.
- Urgent symptoms need medical care, not a quiz or an online diagnosis.

Where next?

Continue to Topic 03 under Understanding Alcohol & Drugs, or open “Get Help & Support Someone” at any point. The website should not require registration, scores, or personal disclosure to access support.

10 | Publication design and review standards

Recommended digital lesson sequence

Element	Website behavior
Introduction	A short question and learning goals; skip-to-support available.
Core concept	One plain-language explanation per screen with optional detail.
Visual diagram	Brain/body systems simplified; text alternative carries the full meaning.
Scenario	Fictional adult/youth examples, not personal-use prompts.
Interactive question	Keyboard accessible options, explanation on selection, retry and skip.
Media	Captioned videos; transcripts and low-data equivalent text.
Help access	Persistent route to approved professional and emergency information.

Professional review gate

- Clinical reviewer: effects, warning signs, medicine interactions, overdose and withdrawal wording.
- Youth safeguarding reviewer: non-stigmatizing scenarios and age-appropriate language.
- Rwanda policy reviewer: verify any local health-service link or emergency contact prior to publication.
- Accessibility reviewer: screen-reader labels, contrast, captions, navigation, reduced motion.
- Language reviewers: medically appropriate Kinyarwanda and French versions.

Data protection: Public learning remains anonymous by default. Do not retain a user’s answers or infer a diagnosis from educational scenarios. Editorial signoff must be recorded separately from publication status.

11 | Scientific and policy reference register

[1] WHO. Alcohol (28 June 2024).

https://www.who.int/news-room/fact-sheets/detail/alcohol

[2] NIAAA. Alcohol and the Brain: An Overview.

https://www.niaaa.nih.gov/publications/alcohol-and-brain-overview

[3] National Institute on Drug Abuse. Drugs, Brains, and Behavior (science overview).

https://nida.nih.gov/research-topics/addiction-science/drugs-brain-behavior-science-of-addiction

[4] NIH Health. Substance Use Disorders.

https://health.nih.gov/health-topics-a-z/substance-use-disorder

[5] NIAAA. Alcohol and the Human Body (March 2026).

https://www.niaaa.nih.gov/alcohols-effects-health/alcohol-topics-z/alcohol-facts-and-statistics/alcohol-and-human-body

[6] NIAAA. Alcohol and the Adolescent Brain.

https://www.niaaa.nih.gov/publications/alcohol-and-adolescent-brain

[7] CDC. What to Do If You Think Someone Is Overdosing.

https://www.cdc.gov/stop-overdose/response/

[8] Rwanda Ministry of Health. Health Sector Strategic Plan V (2024/25–2028/29).

https://www.moh.gov.rw/strategic-plans-priorities

Evidence interpretation

The scientific sources support general educational claims; the Rwanda Health Sector Strategic Plan is a policy-context reference, not an endorsement or national certification of this material. Health claims and legal/clinical applicability must be rechecked at final professional approval.

Editorial record

Prepared 11 November 2026. Status: evidence-informed educational draft. Clinical reviewer: pending. Youth safeguarding reviewer: pending. Language validation: pending. Service verification: pending. Publication approval: not granted.